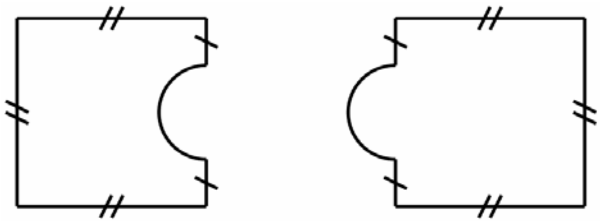
Perimeters and areas of familiar 2D shapes

Teacher resource

Question 1



A jigsaw has pieces of two shapes. Both shapes are based on identical squares and semicircles.

Indicate which statement is true about the jigsaw pieces.

- ⬭ They have identical areas and identical perimeters.
- ⬭ They have different areas but identical perimeters.
- ⬭ They have identical areas but different perimeters.
- ⬭ They have different areas and different perimeters.

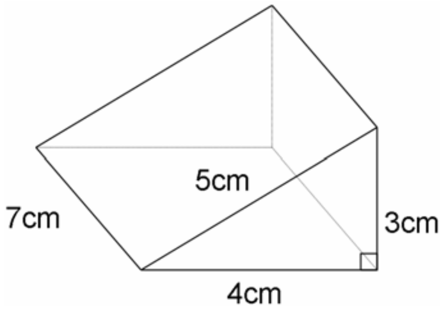
Answer: B
Skill: Students understand the difference between perimeter and area.

Additional questions

1. If the diameter of the semicircle is 1 cm and the length of the square is 2 cm. calculate the perimeters of the two different shaped jigsaw pieces.
2. In a jigsaw with larger pieces, the square on which the pieces are based has an area of 9 square centimetres. The semi-circle has a diameter of 1.5 cm. What is the perimeter of these pieces?

Question 2

This diagram shows a triangular prism and its dimensions.



Which statement shows the surface area of this prism in square centimetres?

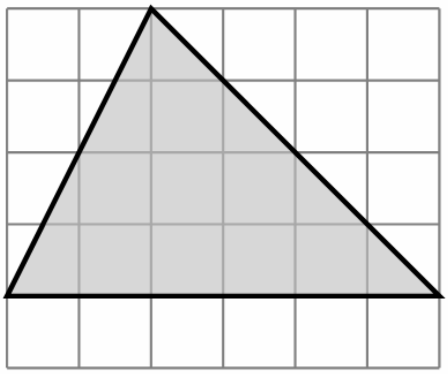
- (3 x 4 x 5) + (3 x 4 x 5) + (5 x 7) + (3 x 7) + (4 x 7)
- (3 x 4) + (12 x 7)
- (3 x 4) + (3 x 4) + (5 x 7) + (3 x 7) + (4 x 7)
- (3 x 4) + (3 x 4) + (12 x 7)

Answer: B
Skill: Students calculate the surface area of a triangular prism.

Additional questions

1. How would statement B change if the height of the prism was doubled?
2. Would the surface area of the prism double if the height was doubled?
3. Would the volume of the prism double if the height of the prism was doubled?
4. Two of the triangular prisms shown in the diagram are used to make a rectangular prism. Change one number in Statement D to ensure the statement shows the surface area of this rectangular prism.

Question 3



Each square on this grid has an area of 10 square centimetres.
What is the area of this triangle in square centimetres?

______________ cm^2

Answer: 120 cm^2
Skill: Students calculate the area of a triangle

Background information

Students should understand the area of a triangle as half the area of a rectangle and that the formula *area of a triangle = half base x perpendicular height* is an abbreviated expression of this relationship.

Additional questions

1. If the area of one square on the grid is 0.5 cm^2, what is the area of the triangle?
2. If the area of the triangle is 5400 mm^2, what is the area of one square on the grid in square centimetres?

Question 4

The length of the diagonal of a square of paper is 20cm.
The square is cut along both diagonals to form four identical triangles.

What is the area of one of the triangles?

_______________ cm^2

Answer: 50 cm^2
Skill: Students use information given in context to calculate the area of a triangle.

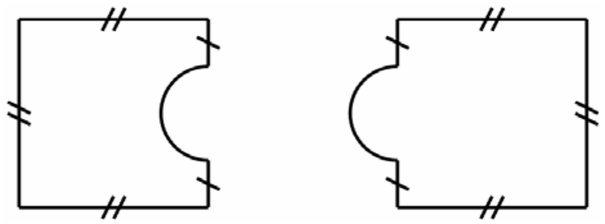
Background information
This question is an example of a two-step problem. Students may need support to understand that the information needed to solve the problem is not specifically stated in the question. In many situations, representation of the information given as a diagram supports the process of finding a solution.

Additional questions

1. A square of paper is cut along both its diagonals to form four identical triangles. The area of one triangle is 20 cm^2. What is the length of the diagonal of the square?
2. A square has 20 cm sides. What is the length of the diagonals, correct to two decimal places?

Student activity: Perimeters and areas of familiar 2D shapes

Question 1



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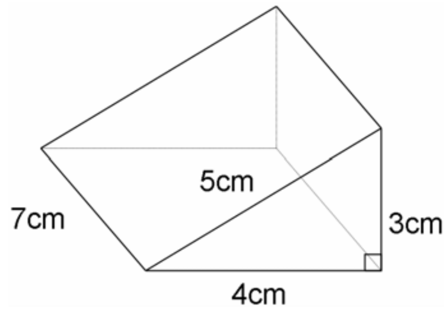
Question 2

This diagram shows a triangular prism and its dimensions.



Which statement shows the surface area of this prism in square centimetres?

- ⬭ (3 x 4 x 5) + (3 x 4 x 5) + (5 x 7) + (3 x 7) + (4 x 7)
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- ⬭ (3 x 4) + (3 x 4) + (5 x 7) + (3 x 7) + (4 x 7)
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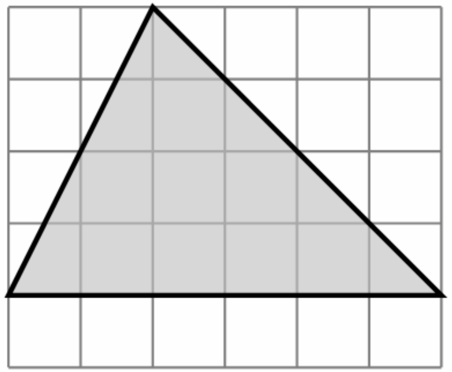
Reviewed January 2014

Question 3



Each square on this grid has an area of 10 square centimetres.
What is the area of this triangle in square centimetres?

_____________ cm^2

Question 4

The length of the diagonal of a square of paper is 20cm. The square is cut along both diagonals to form four identical triangles.

What is the area of one of the triangles?

_____________ cm^2

Reviewed January 2014